

Luc A. Barrett

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Experimental physics Ph.D. student interested in quantum information, particle physics, cosmology, and machine learning

Education



Cornell University
DOCTOR OF PHILOSOPHY IN APPLIED PHYSICS

Ithaca, NY
Starting Fall 2025



University of Massachusetts, Amherst
MASTER OF SCIENCE IN COMPUTER SCIENCE

Amherst, MA
In-Progress, Expected May 2025

Bay State Fellow
GPA: 3.9
Relevant Courses: Reinforcement Learning, Machine Learning, Quantum Communication, Advanced Robotics



University of Massachusetts, Amherst
BACHELORS OF SCIENCE IN PHYSICS, *Cum Laude*

Amherst, MA
2024

Commonwealth Honors Scholar with Greatest Distinction
GPA: 3.94
Thesis: “Characterization of Drifting Charge Clusters in Liquid Xenon Produced by a Laser-Driven Photocathode for the nEXO Experiment”,
Advised by Prof. Krishna Kumar
Relevant Courses: Physical Quantum Computation, (Graduate) Classical Mechanics, Quantum Mechanics



University of Massachusetts, Amherst
BACHELORS OF SCIENCE IN MATHEMATICS, *Cum Laude*

Amherst, MA
2024

Relevant Courses: Nonlinear Dynamics & Chaos, Differential Geometry, Real Analysis, PDEs



University of Massachusetts, Amherst
BACHELORS OF SCIENCE IN COMPUTER SCIENCE, *Cum Laude*

Amherst, MA
2024

Relevant Courses: Intro. to Machine Learning, Algorithms, NLP, Artificial Intelligence, Quantum Algorithms

Research Experience



UMass Amherst College of Information & Computer Sciences
MASTER’S PROJECT

Amherst, MA
2024 - 2025

- Apply modern reinforcement learning algorithms to optimize link generation between quantum memory registers with various noise models, and minimal simplifying assumptions

Simulation Quantum Information



UMass Amherst
UNDERGRADUATE RESEARCH

Amherst, MA
2021 - 2024

- Worked on the MOLLER experiment, developing a software tool to check ‘two-bounce’ conditions. Validates the design of the collimator system and ensures no accidental avenues for noise
 - Used Rust, multithreading, and optimized collision check algorithms to bring simulation time from several days to a couple of minutes. Uses geometry directly imported from Geant4.
 - Presented in APS DNP CEU undergraduate poster session
- Worked on the nEXO neutrinoless double beta decay ($0\nu\beta\beta$) experiment
 - Under Prof. Krishna Kumar, worked on characterization of charge clusters drifting in liquid xenon produced by a laser-driven photocathode
 - Work from the team presented in poster session at Neutrino 2024
 - Under Prof. Andrea Pocar, worked on studies using experimental data and GPU-accelerated Chroma simulations to study optical properties of materials in LXe, at the scintillation wavelength of Xe
 - Work presented in internal nEXO Collaboration meeting
- Started development of a Julia simulation package to simulate gaussian quantum states. Implement efficient representations and operations to study these states, with applications to quantum sensing and algorithms

Nuclear Physics $0\nu\beta\beta$ Algorithm Design Simulation Quantum Information

Posters & Presentations

- [1] L. Barrett, *Chroma Optics Measurements* (nEXO Light Simulation Workshop, 2024).
- [2] L. B. K. S. K. M. L. A. N. J. Z. Devin Cesmecioglu Jason Bane, *Photo-Induced Charge Calbration for Nexo* (Neutrino 2024, 2024).
- [3] K. S. K. Luc Barrett Jon Mott, *Program to Identify Secondary Background Sources in the MOLLER Experiment* (APS DNP & JPS Fall Meeting, 2023).

Awards & Honors

- 2025 **Fulbright Semifinalist**, Fulbright Association
- 2024 **Bay State Fellowship**, UMass Amherst College of Information & Computer Sciences
- 2024 **Kandula Sastry Undergraduate Award**, UMass Amherst Department of Physics
- 2024 **Phi Beta Kappa**, PBK Society
- 2023 **LeRoy F. Cook Jr. Memorial Award**, UMass Amherst Department of Physics
- 2023 **Phi Kappa Phi**, PKP Society
- 2020-2024 **Dean's List (x7, all full-time semesters)**, UMass Amherst

Teaching Experience



UMass Amherst College of Information & Computer Sciences
GRADUATE TEACHING ASSISTANT

Amherst, MA
2024 - 2025

- CS648: Quantum Information Systems. For the class of ~40 graduate students, I held weekly office hours, and additional review lectures to cover optional practice problem sets I designed. Graded and proctored midterm & final exams
- CS250: Introduction to Computation. For the class of ~300 undergraduate students, I held weekly office hours, proctored exams, and graded weekly homework assignments.



UMass Amherst
UNDERGRADUATE TEACHING ASSISTANT

Amherst, MA
2024 - 2025

- CS490Q: Quantum Information Science. Held in-person office hours and exam review. Graded homework assignments
- PHYS281: Computational Physics (x2). Assisted students with in-class exercises and held office hours. Held optional supplemental lectures to cover additional relevant content (OOP, data structures, API's, multiprocessing, cluster computing).
- PHYS181: Introduction to Mechanics. Assisted students with assignments in-class, held exam-review and office hours.

Mentoring Experience



UMass Amherst
UNDERGRADUATE PEER MENTOR

Amherst, MA
2022 - 2024

- Invited by both physics and CS department faculty to be paired with incoming freshman students as a peer mentor. Met regularly to provide social and academic support, particularly when involving course selection/planning, and getting involved in research

Skills

Toolkits	NumPy Pandas Geant4 ROOT Qiskit MatLab Mathematica
Programming	Python Julia Rust C/C++ Javascript/Typescript Java SQL
Personal Interests	Guitar/Bass Reading 3D Printing